Computational Motor Control (MT-436)

Amphibious Locomotion of Polymander

A Salamander-like Robot

Project #1

Group	13
Guilhem Destriau	343728
Medina Filomena Colabrese	406678
Romain Corbel	344198

Prof. Auke Ijspeert

Table of Contents

1 Wave Controller and Metrics **3**
1.1 Qst 1.1 : Wave controller implementation . . . 3
1.2 Qst 1.2 : Metrics implementation . . . 3
1.3 Qst 1.3 : Parameter exploration . . . 4

2 CPG controller **5**
2.1 Qst 2.1 : Controller Implementation and movement analysis . . . 5
2.2 Qst 2.2 : Parameter exploration . . . 6
2.3 Qst 2.3 : Animal comparison . . . 7
2.4 Bonus question . . . 7

3 Sensory feedback **8**
3.1 Qst 3.1 : Stretch Feedback . . . 8
3.2 Qst 3.2 : Varying stretch feedback strength . . . 9
3.3 Qst 3.3 : Neural disruptions . . . 10

4 Appendix and Bonus Question **11**

5 Generative AI declaration **13**

1 Wave Controller and Metrics

1.1 Qst 1.1 : Wave controller implementation

To translate the wave equation into left ($M_{L,i}$) and right ($M_{R,i}$) muscle activations, we use :

$$M_{L,i}(t) = \max(\text{wave}_i(t), 0), \quad M_{R,i}(t) = \max(-\text{wave}_i(t), 0) \tag{1}$$

This ensures that only one side of the joint is active at a time to create the required bending torque.

The implemented wave controller can be seen in the video : exercise1_1_Wave_Controller.mp4.

FIGURE 1 – Snapshot of the *Polymander* swimming simulation using the Wave Controller with parameters $A = 2$, $f = 2$ Hz, and $TWL = 1$. The traveling wave propagates from head to tail, producing forward propulsion.

1.2 Qst 1.2 : Metrics implementation

We implement the given performance metrics [1] for the wave controller ($A = 2.0, f = 2.0$ Hz, $TWL = 1.0$) and present them in Table 1 :

ID	Metric	Value
NM1	Mean Neural Frequency	1.9992 Hz
	Mean Neural Amplitude	1.9997
NM2	Mean Intersegmental Phase Lag (IPL_{neur})	0.7752 rad
MM1	Mean Mechanical Frequency	1.9992 Hz
	Mean Mechanical Amplitude	0.4549
MM2	Forward Speed (v_{fwd})	0.1376 m/s
	Lateral Speed (v_{lat})	-2.17×10^{-4} m/s
MM3	CoM Displacement	1.3412 m
MM4	Mechanical Energy (E)	0.1120 J
	Cost of Transport (CoT)	0.0709 J/m

TABLE 1 – Locomotion performance metrics for the baseline wave controller, categorized by Neural (NM) and Mechanical (MM) criteria.

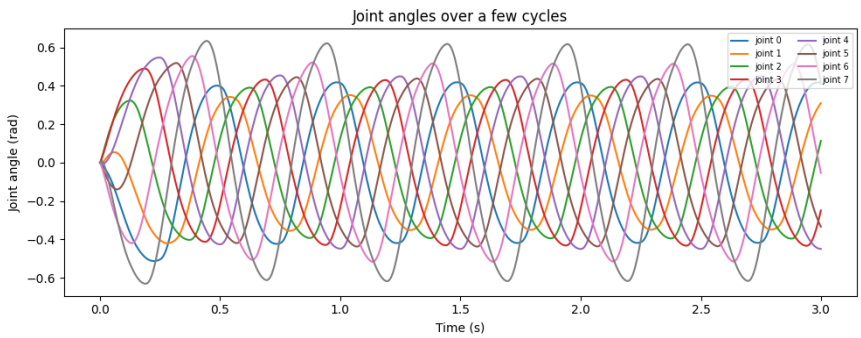


FIGURE 2 – Time evolution of joint angles.

The simulation results in Table 1 confirm that the wave controller matches our expectations. Indeed both the neural and mechanical frequencies are equal to 1.9992 Hz and the neural amplitude is 1.9997, closely matching the controller configuration (A = 2.0, f = 2.0 Hz).

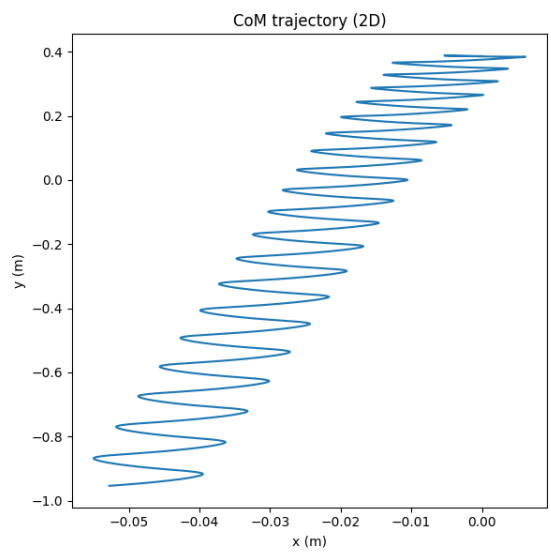


FIGURE 3 – 2D trajectory of the Center of Mass highlighting the periodic lateral oscillations associated with undulatory swimming.

Furthermore, Figure 3 shows the COM trajectory. This trajectory confirms that the robot swims in an essentially straight line along the y-axis, covering around 1.4 m in 5 s while lateral oscillations in x remain negligibly small. We can also see a small drift in the x-direction.

1.3 Qst 1.3 : Parameter exploration

To investigate the influence of kinematic parameters on swimming performance, a 2D grid search was performed over the amplitude $A \in [1.0, 4.0]$ and total wave lag $TWL \in [0.2, 1.5]$ at a fixed frequency of $f = 1.5$ Hz.

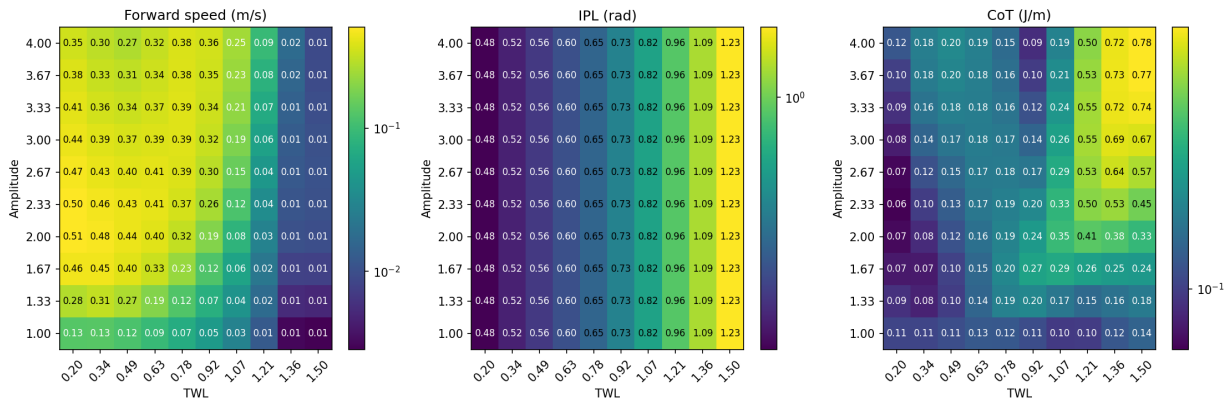


FIGURE 4 – Heatmaps of forward speed v_{fwd} (left) and Cost of Transport CoT (right) across the A-TWL parameter space.

We observe in Figure 4 that forward speed increases with higher A and lower TWL, while IPL_{neur} increases with TWL and is independent of A; finally, both TWL and A increases contribute to an increase in CoT.

2 CPG controller

2.1 Qst 2.1 : Controller Implementation and movement analysis

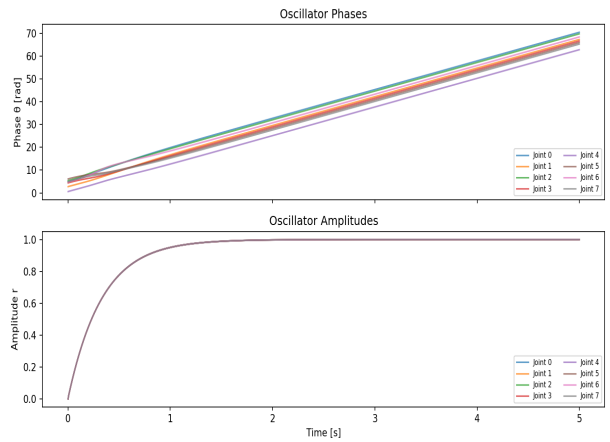
The CPG controller is implemented with a fixed bilateral drive $d = 3$ yielding $f_i = 2\,\text{Hz}$ and $R_i = 1.0$. The controller can be seen in the video : exercise2_1_CPG controller_without_Sensory_Feedback.mp4

Figure 5a shows that the phases θ_i grow linearly at a rate of approximately $2\pi \times 2\,\text{rad}\,\text{s}^{-1} = 4\pi\,\text{rad}\,\text{s}^{-1}$ (phase increases of approximately 12 rad in 1 s). We also see that amplitudes converge exponentially to 1.

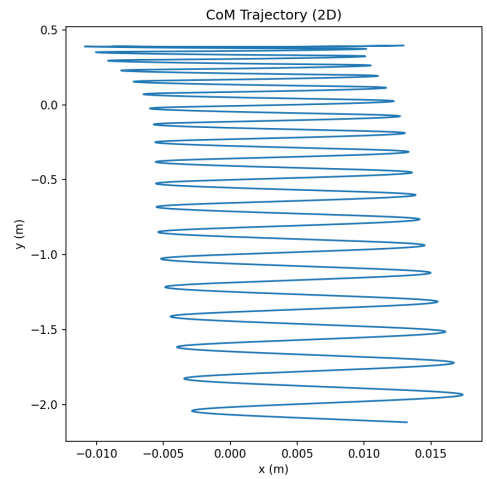
All 8 joints oscillate at the same frequency ($f \approx 2$ Hz), producing the characteristic rostro-caudal travelling wave visible in figure 5d (we applied vertical offsets for clarity). After a short transient ($t \approx 0.5$ s), all joints settle into regular sinusoidal oscillations with uniform peak-to-peak amplitude and stable phase relationships, forming a coherent undulatory wave propagating from the rostral (joint 0) to the caudal (joint 7) end of the body.

The muscle output sum $M_L + M_R$, shown on figure 5c, converges to the constant value of 2.0 for all joints after a brief transient ($t \lesssim 1$ s). The difference $M_L - M_R$ settles into stable sinusoidal oscillations of amplitude ~2.0 at $f \approx 2$ Hz.

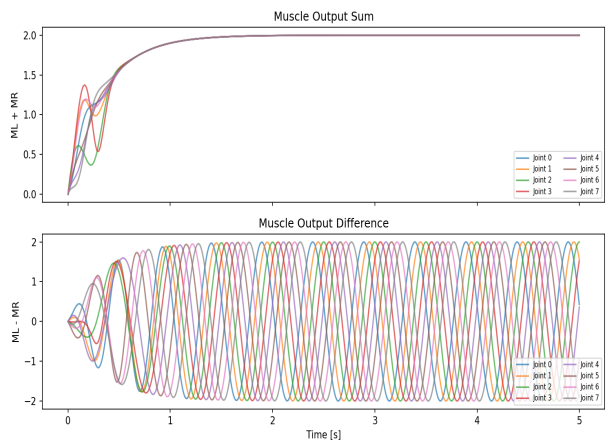
The CoM trajectory figure 5b confirms that the robot swims in an essentially straight line along the y-axis, covering ~2.1 m in 5 s while lateral oscillations in x remain negligibly small (≤0.03 m peak-to-peak). A slight lateral drift is nonetheless visible, with the envelope of x-oscillations shifting between ~0 m and ~0.007 m.



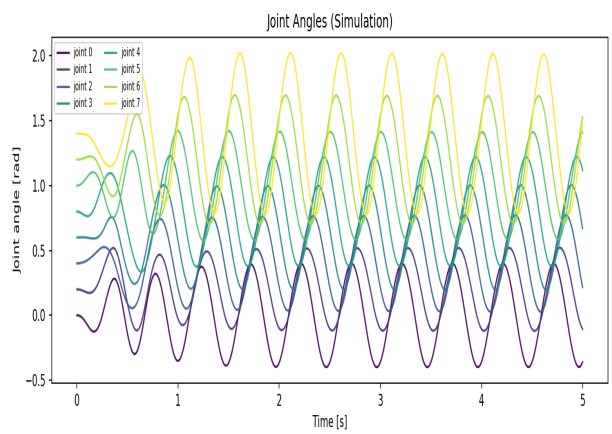
(a) Oscillator States (θ, r)



(b) 2D CoM trajectory



(c) Muscle Output Sum/Diff



(d) Joint angles (8 joints)

FIGURE 5 – Neural and physical dynamics over 5s : (a) oscillator internal states, (b) center of mass trajectory, (c) muscle outputs, and (d) joint angles.

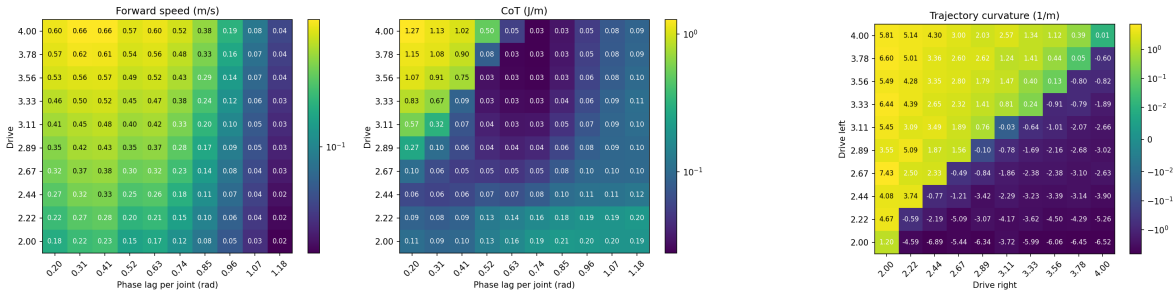
2.2 Qst 2.2 : Parameter exploration

Having validated the CPG controller in the previous section, we now investigate how two key parameters, the bilateral drive $d = d_{\text{left}} = d_{\text{right}}$ and the total body phase bias $\varphi_{\text{body_total}}$, affect locomotion performance. We then study the effect of a differential drive $d_{\text{left}} \neq d_{\text{right}}$ on the robot trajectory curvature.

Forward speed increases with drive d and decreases with phase lag PL, peaking at ~0.66 m/s for $d = 4.0$ and $PL \approx 0.31$ rad/joint, while CoT is lowest (~0.03 J/m) for intermediate drives ($d \in [3.1, 3.8]$) and small phase lags (P L < 0.52 rad/joint).

For the differential drive, the curvature heatmap shows a clear antisymmetric structure along the diagonal : when $d_{\text{left}} > d_{\text{right}}$ the robot turns right (negative curvature) and vice versa, with

curvature magnitude growing with the drive asymmetry $|d_{\text{left}} - d_{\text{right}}|$. Straight locomotion ($\kappa \approx 0$) is recovered along the diagonal $d_{\text{left}} = d_{\text{right}}$.



(a) Forward speed and CoT.

(b) Trajectory Curvature.

FIGURE 6 – Performance metrics and maneuverability analysis : (a) Forward speed and CoT heatmaps over bilateral drive d and phase lag PL; (b) Trajectory curvature over differential drive inputs $d_{\text{left}}, d_{\text{right}}$.

2.3 Qst 2.3 : Animal comparison

To compare animal and controller locomotion, we extracted three metrics from the recorded joint kinematics : global oscillation frequency, per-joint angular amplitude, and inter-joint phase lag. The results are shown in the Figure 7.

We see that the animal swims at ~1.55 Hz while the controller runs at 2.0 Hz, a ~29% overestimate. Amplitude profiles both increase from head to tail (J0–J7), but the animal shows lower amplitudes at head joints (J0 : ~0.10 rad vs ~0.35 rad for the controller) and comparable values at tail joints (~0.60 rad vs ~0.55 rad). Inter-joint phase lags are more variable in the animal (0.40–1.45 rad) than in the controller (0.63–0.80 rad), with mean values of 0.842 rad (animal) and 0.689 rad (controller), a ~18% underestimate.

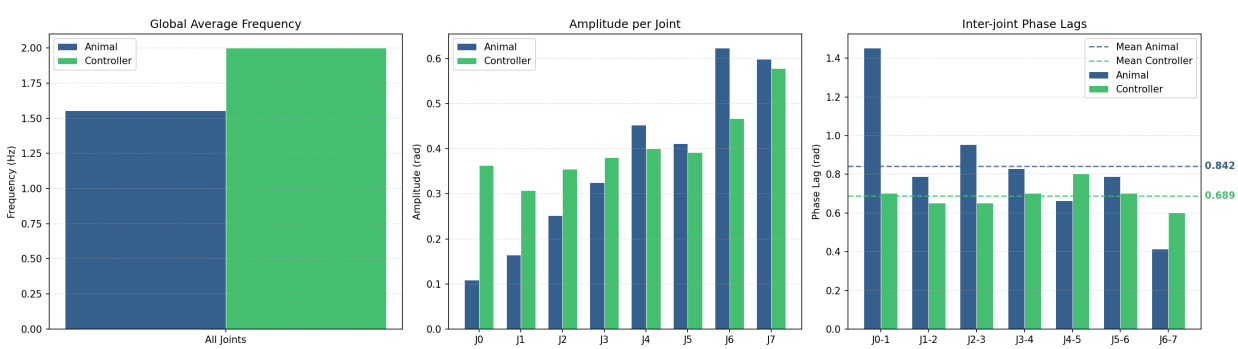


FIGURE 7 – Comparison between animal data (blue) and CPG controller (green) for global frequency (left), amplitude per joint (centre), and inter-joint phase lags (right).

2.4 Bonus question

The bonus question is done in the appendix.

3 Sensory feedback

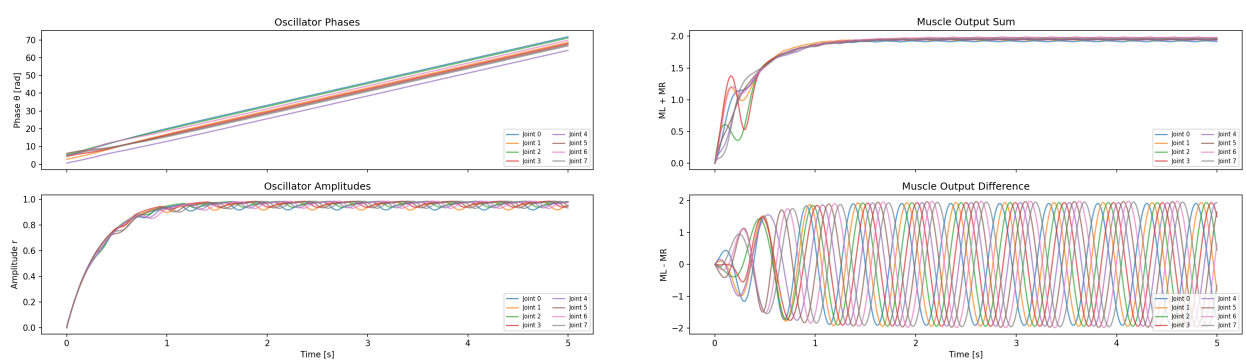
3.1 Qst 3.1 : Stretch Feedback

Stretch feedback was added to the CPG as described in equations (12) to (18) in the project manual [1]. The controller can be seen in the video : animation3_1_with_Sensory_Feedback.mp4.

Varying w^{ipsi} is significant, as this is the value that directly determines the contribution of stretch feedback to the overall system. If $w^{ipsi} = 0$ then the stretch feedback terms are 0, and thus stretch sensor readings are ignored. If w^{ipsi} is very large, then the contribution from stretch feedback will be more significant than the original oscillator equations, and the system will be primarily relying on the joint sensors as opposed to the oscillator intrinsics.

The simulation was run with $w^{ipsi} = 3.0$ and $w^{ipsi} = 0.0$, with a runtime and buffer size of 20001. Figure 8 shows the time evolution of amplitude and gain, and the sum and difference of muscle outputs, for comparison with the results from Section 2.1. Figure 9 shows the comparison of trajectories and body joint angles for the first five seconds of the trial for both cases. Table 2 shows the metrics for both cases.

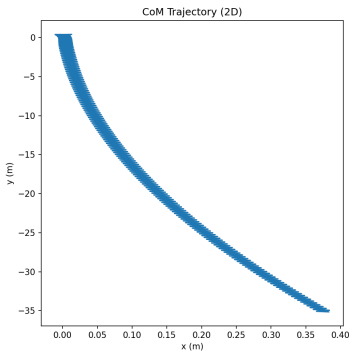
Adding sensory feedback slightly increased the neural frequencies and slightly decreased neural amplitudes. It also caused the oscillator amplitudes to vary between around 0.85 and 1, instead of remaining constant as they were before. The result is that the robot's forward speed and distance traveled were increased, and cost of transport was decreased. In other words, adding sensory feedback increased the robot's efficiency, allowing it to produce better physical results, as it allows the robot to react to its position in its environment instead of relying solely on internal oscillators.



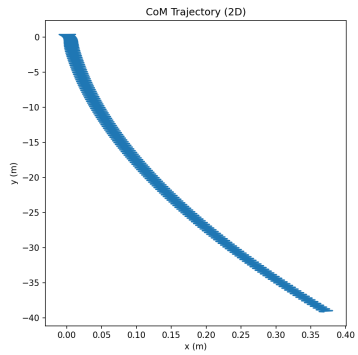
(a) Oscillator States (θ, r)

(b) Muscle Output Sum and Difference

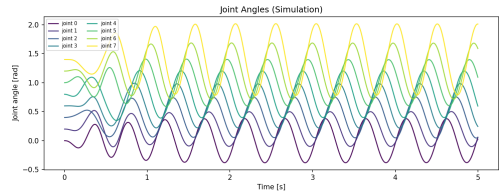
FIGURE 8 – Neural dynamics over the first 5s (with $w^{ipsi=3.0}$)



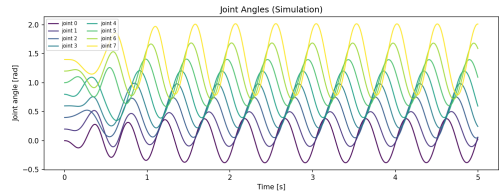
(a) COM trajectory – $w^{ipsi} = 0.0$



(b) COM trajectory – $w^{ipsi} = 3.0$



(c) Joint angles – $w^{ipsi} = 0.0$



(d) Joint angles – $w^{ipsi} = 3.0$

FIGURE 9 – CoM trajectory and joint angles comparison

Metric	$w^{ipsi} = 0$	$w^{ipsi} = 3.0$
Neural frequencies [Hz]	1.9999	2.0480
Neural amplitudes [-]	2.2512, 2.0985, 1.9030, 1.7638 1.7650, 1.9095, 2.1082, 2.2664	2.2417, 2.0635, 1.8047, 1.6165 1.6053, 1.8049, 2.0644, 2.2657
Forward speed [m/s]	0.4575	0.5087
CoT [J/m]	0.02367	0.02072

TABLE 2 – Metrics with and without stretch feedback

3.2 Qst 3.2 : Varying stretch feedback strength

Figure 10 shows the impacts of varying w^{ipsi} on cost of transport, forward speed, neural frequency, and neural amplitude. Larger w^{ipsi} values consistently produce larger forward speeds and neural frequencies, however CoT and neural amplitudes both show a minimum around $w^{ipsi} = 12$ before these metrics begin to increase again (with CoT increasing quite sharply). This shows that, although the speed may continously increase, it becomes significantly more expensive past this threshold, with this point marking the robot's peak efficiency.

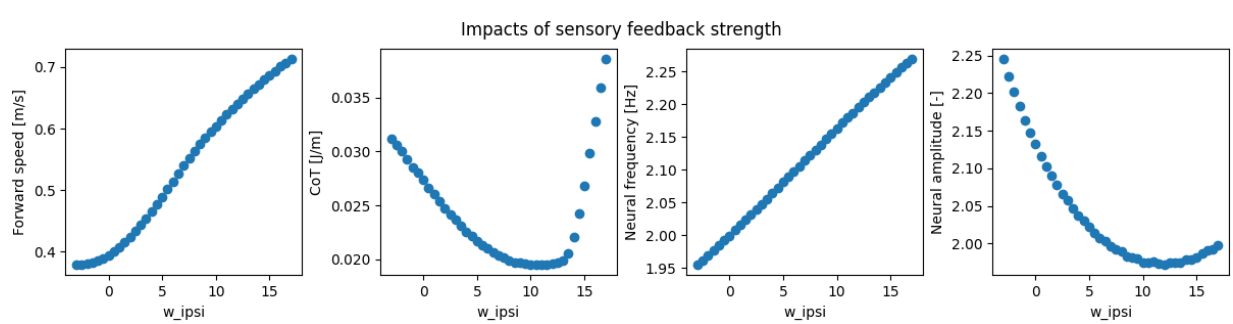
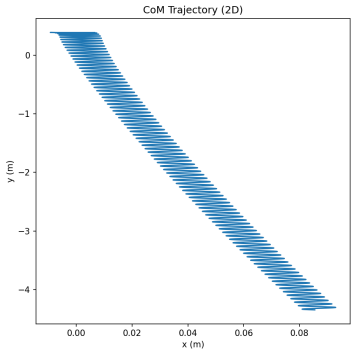


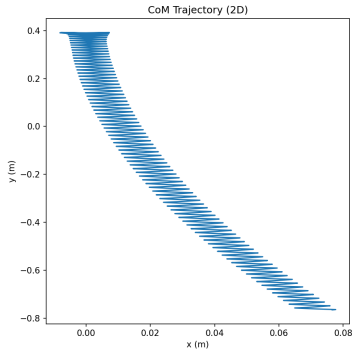
FIGURE 10 – Impacts of varying w^{ipsi}

3.3 Qst 3.3 : Neural disruptions

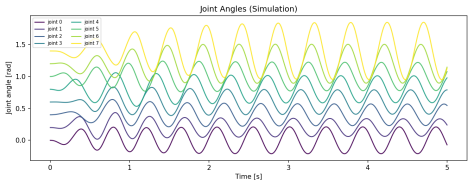
Figure 11 show the center of mass trajectories and joint angles compared with and without a 20% disruption. The robot travels over four times farther in the case with no disruptions, indicating that a 20% disruption in both sensory feedback and couplings is enough to severely impair its motion. Observing the simulation, the robot's wavelike motion appears much less smooth and consistent when the disruptions are present, it seems like the it is struggling to converge to a reliable, consistent motion pattern. The videos with and without disruption are provided in animation_3_3_with_disruptions.mp4 and animation_3_3_no_disruptions.mp4, respectively.



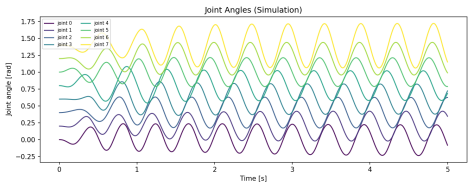
(a) COM trajectory – no disruptions



(b) COM trajectory – 20% disruptions



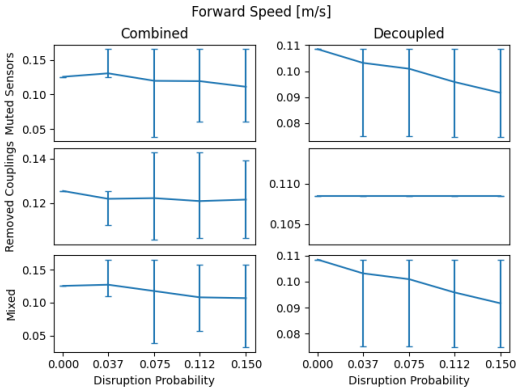
(c) Joint angles – no disruptions



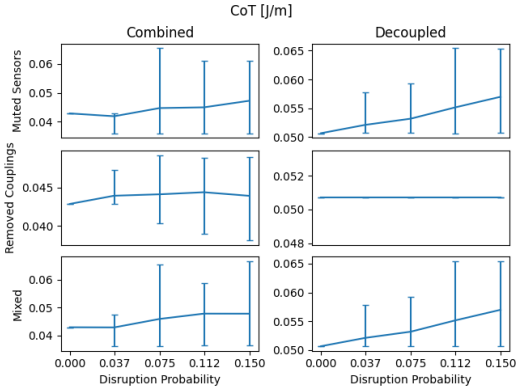
(d) Joint angles – 20% disruptions

FIGURE 11 – CoM trajectory and joint angles comparison

Figure 12 shows the comparison of forward speed and cost of transport for the six disruption and controller setup cases. The means are plotted over 8 random cases with maximum and minimum indicated. The forward speed decreases as more disruptions are added (matching the observations from Figure 11) and the cost of transport increases, indicating reduced efficiency. These trends are much more significant for the decoupled controller as the robot is operating with less information than in the combined case (however the "removed couplings" disruption has no impact on the decoupled controller, as all couplings are already removed).



(a) Forward speed



(b) $w^{ipsi} = 3.0$

FIGURE 12 – Cost of transport

4 Appendix and Bonus Question

Reproducibility

All results in this report are reproducible by running the individual Python files (e.g., exercise1_1.py, exercise1_2.py, etc.) or by executing project1.py that will automatically run all of the exercises (except the bonus that needs to be run individually). Note that all plots are saved to the logs folder and will only be displayed if show_plot is set to True in exercise_all.py.

Bonus : Controller Tuning to Imitate Animal Swimming

Objective and Loss Function

The goal is to tune the CPG controller parameters to match three key kinematic metrics extracted from animal swimming data : the mean oscillation frequency f_{animal}, the mean joint amplitude A_{animal}, and the mean inter-joint phase lag ϕ_{animal}. The scalar loss is defined as the sum of squared relative errors :

$$\mathscr{L}(\theta) = \left(\frac{f_{\text{sim}} - f_{\text{animal}}}{f_{\text{animal}}}\right)^2 + \left(\frac{A_{\text{sim}} - A_{\text{animal}}}{A_{\text{animal}}}\right)^2 + \left(\frac{\phi_{\text{sim}} - \phi_{\text{animal}}}{\phi_{\text{animal}}}\right)^2 \tag{2}$$

with target values $f_{\text{animal}} = 1.553$ Hz, $A_{\text{animal}} = 0.351$ rad, $\phi_{\text{animal}} = 0.842$ rad.

Optimized Parameters

Four scalar parameters were selected for optimization, as they jointly control frequency, amplitude, and phase lag :

Parameter	Role	Initial value (Exercise 2.1)	Optimized value
d	Drive	3.000	2.736
PL	Phase lag	0.500	0.970
`offset_freq`	Offset of the oscillator frequency in Hz	1.000	0.445
G_{freq}	Gain of the drive-dependent frequency per oscillator	0.500	0.732

Optimization Method : Batch Gradient Descent

Gradient descent was performed with numerical gradients estimated via central finite differences ($h = 0.05$). Each gradient step requires $2 \times 4 = 8$ simulations, organized in two sequential batches of 4 parallel workers :

- **Batch 1** : perturb d and ϕ_L ($\pm h$) $\rightarrow$ estimate $\partial\mathscr{L}/\partial d$ and $\partial\mathscr{L}/\partial\phi_L$.
- **Batch 2** : perturb `offset_freq` and G_{freq} ($\pm h$) $\rightarrow$ estimate $\partial\mathscr{L}/\partial$`offset_freq` and $\partial\mathscr{L}/\partial G_{\text{freq}}$.

The parameter update with learning rate $\eta = 0.05$ and gradient norm clipping reads :

$$\theta_{t+1} = \text{clip}\left(\theta_t - \eta \frac{\nabla_\theta \mathscr{L}}{\max(1, \|\nabla_\theta \mathscr{L}\|)}, \theta_{\min}, \theta_{\max}\right) \tag{3}$$

The optimization ran for 50 iterations, reducing the loss from $\mathscr{L}_0 = 0.284$ to $\mathscr{L}_{50} = 0.036$ (87 % reduction).

Convergence Analysis

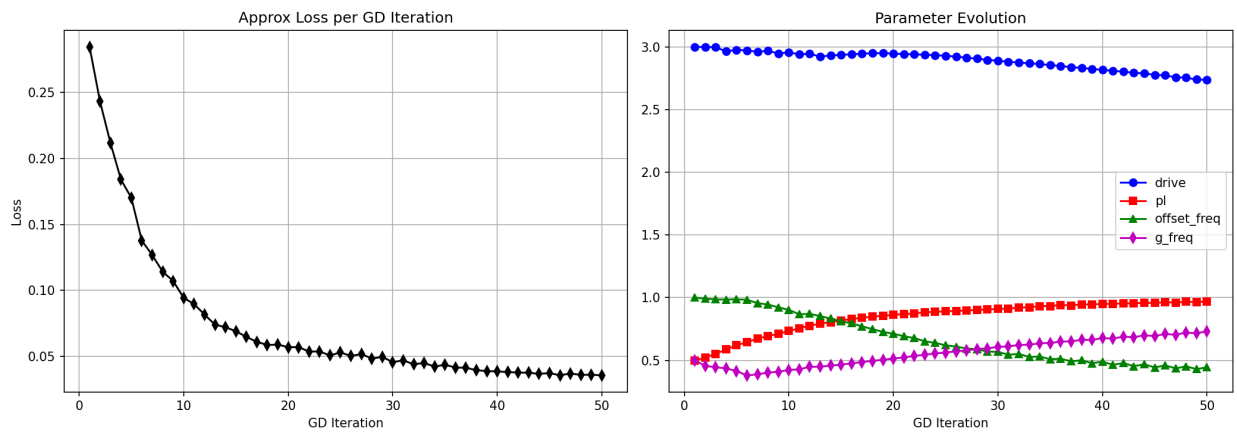


FIGURE 13 – Loss over iteration and Evolution of parameter's values over iteration

Figure 13 shows the loss curve and the parameter trajectories over the 50 gradient descent iterations.

Loss (left panel). The loss drops sharply in the first 15 iterations (from 0.284 to ~ 0.05), then converges smoothly to a plateau at 0.036.

Parameter trajectories (right panel). The four parameters evolve very differently, which reveals which degrees of freedom are most important for matching the animal :

- ϕ_L (`pl`, red) : the largest relative change, increasing from 0.50 to 0.97 rad (+94%).
- `offset_freq` (green) : decreases from 1.00 to 0.44 (−56%).
- G_{freq} (purple) : increases moderately from 0.50 to 0.73 (+46%).
- d (`drive`, blue) : barely moves (3.00 → 2.74, −9%).

The smooth, monotone convergence of all four parameters confirms that the central-difference gradient estimates are sufficiently accurate.

Results

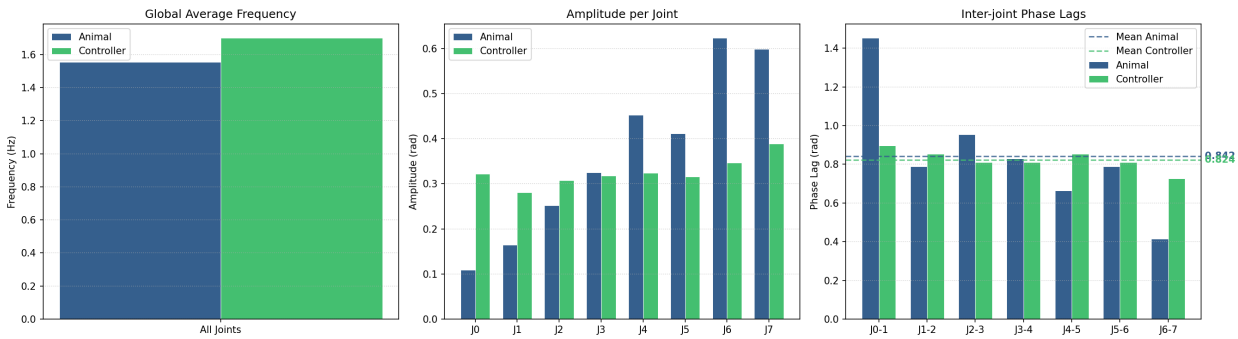


FIGURE 14 – Comparison between animal data (blue) and the optimized CPG controller (green) for global frequency (left), amplitude per joint (centre), and inter-joint phase lags (right).

Figure 14 compares the animal and optimized controller kinematics.

- **Frequency** : the controller reached 1.68 Hz against the animal target of 1.55 Hz, a relative error of **8**%.
- **Inter-joint phase lag** : the mean controller IPL converged to 0.842 rad, very close to the animal.
- **Amplitude** : the overall amplitude profile is close to the animal, although the rostro-caudal gradient (joints J0–J7 increasing in the animal) remains difficult to capture with a uniform G_{amp}.

The residual frequency error is attributed to the constraint that `offset_freq` and G_{freq} are uniform across all eight joints; per-joint values would provide additional degrees of freedom to further reduce the loss.

5 Generative AI declaration

Generative AI tools have been used in the development of this report and its associated code to assist with code quality, structure, and idea clarification. All outputs provided by such tools were critically reviewed, validated, and adapted as necessary. The authors take full responsibility for the final content, and all core ideas, implementations, and conclusions are their own.

Références

[1] Computational Motor Control teaching team. Amphibious locomotion of polymander : A salamander-like robot, 2026.